

SAPIENZA
UNIVERSITÀ DI ROMA

Space and astronautical engineering

Homework 1

Lyapunov orbits

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Contents

1 Introduction **2**

2 Development **3**

2.1 Dynamical model and equilibrium points 3

2.2 Linear analysis and first guess state 4

2.3 Single Shooting correction 6

2.4 Continuation scheme 8

2.5 Orbit pairs 11

3 Full Algorithm **13**

1 Introduction

This report focuses on the computation and analysis of Lyapunov orbits within the framework of the Planar Circular Restricted Three-Body Problem (PCR3BP), specifically applied to the Earth-Moon system. Lyapunov orbits are defined as planar, symmetric, periodic solutions located around the unstable collinear libration points. These orbits are of significant interest in astrodynamics because, despite their instability, they provide fundamental insights into the natural transport of matter within binary planetary systems.

The phase space structure near these equilibrium points is characterized by a "saddle × center" topology, possessing both stable and unstable manifolds and a center subspace where periodic motion occurs. The objective of this study is to numerically compute the continuous families of Lyapunov orbits associated with the $\mathcal{L}_1$ and $\mathcal{L}_2$ points.

The methodology adopted in this work follows a structured four-step approach:

1. **Analytical Linearization:** We begin by linearizing the Equations of Motion (EOM) around the equilibrium points. By analyzing the eigenvalues and eigenvectors of the Jacobian matrix, we derive the analytical initial conditions for small-amplitude orbits.

2. **Differential Correction (Single Shooting):** Since the linear approximation eventually diverges in the full nonlinear model, we implement a Single Shooting scheme to refine the orbit. This process exploits the symmetry of the PCR3BP, specifically the condition of perpendicular crossings of the x-axis, and utilizes the State Transition Matrix (STM) to correct the initial state until periodicity is achieved.

3. **Family Continuation:** Once a single converged orbit is obtained, we extend the results to compute the entire Lyapunov family. This is achieved using a numerical continuation scheme (Natural Continuation) to step along the solution branch, parameterizing the orbits by their initial position or period.

4. **Bisection for Jacobi Constants:** Finally, we analyze the energetic properties of the computed families. Using a bisection method, we identify and extract specific pairs of orbits from the $\mathcal{L}_1$ and $\mathcal{L}_2$ families that share the exact same Jacobi constant.

2 Development

2.1 Dynamical model and equilibrium points

The problem domain is the Earth-Moon system; the reference frame is rotating (synodic) and is centered on the center of mass of the Earth-Moon system. The problem of Lyapunov orbits is that they are very unstable, so it's crucial to reduce error propagation while performing the computation; for this reason, every variable is normalized, and the positions of the collinear libration points are calculated inside the code and not given as input data.

Table 1: Calculated parameter of the system

Parameter	**Value**
GM_{Erth}	3.986×10^5 km^3/s^2
GM_{Moon}	4.902×10^3 km^3/s^2
Earth-Moon distance (l^*)	384 400 km
Mass parameter (μ)	0.01215
Position L_1	0.8369
Position L_2	1.1557

The goal is to correctly evaluate stable periodic orbits around L1 and L2, two of the three collinear libration points, also called Lagrangian points; L1 is located between the Earth and the Moon, L2 is located next to the moon on the same line, as shown in fig:1

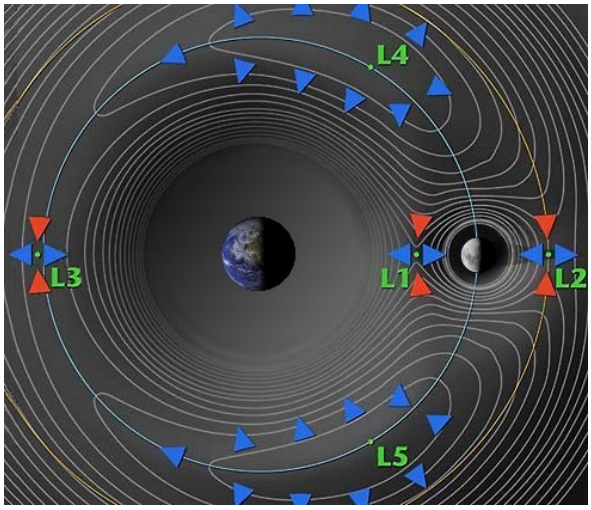


Figure 1: Earth-Moon Lagrange points

The EOM of the system used are the following

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix} = 2 \begin{bmatrix} \dot{y} \\ -\dot{x} \end{bmatrix} + \begin{bmatrix} U_x \\ U_y \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{u} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} u \\ v \\ 2v + U_x \\ -2u + U_y \end{bmatrix}$$

U is defined as the potential, Ux and Uy are its partial derivatives with respect to x and y.

$$U(x, y) = \frac{x^2 + y^2}{2} + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2}$$

L1 and L2 are calculated by imposing the gradient of the potential equal to zero, they correspond to equilibrium points where gravitational force and centrifugal force balance each other. It's crucial to evaluate mathematically these points because they are unstable points (saddle), so using an already available value will lead to inconsistency due to different approximation error, usually related to the value of μ, the mass parameter.

$$\mu = \frac{GM_{Moon}}{GM_{Earth} + GM_{Moon}}$$

To obtain a first guess, the problem is linearized around the equilibrium points, using the Jacobian matrix A, that is very simple due to the fact that the chosen equilibrium points have the y-coordinate equal to zero.

$$A(X) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ U_{xx} & U_{xy} & 0 & 2 \\ U_{yx} & U_{yy} & -2 & 0 \end{bmatrix}$$

Where Uxx indicates the second derivative of U with respect to x, Uxy and Uyx are the mixed derivative, which are equal for symmetry.
This matrix is used to evaluate the evolution of the state transition matrix with the formula

$$d\phi = A(X) \cdot \phi$$

2.2 Linear analysis and first guess state

Analyzing the Jacobian matrix it is immediately apparent that it has 4 eigenvalues, 2 stable and 2 unstable; the stable ones are complex conjugates and they describe

periodic orbits, it's enough to evaluate the positive one and later obtain the complex pair using symmetry.
From the eigenvalues it's possible to calculate the initial state guess, fixing a displacement from the equilibrium point.

$$\omega = \mathrm{Im}(\lambda) \tag{1}$$

$$\bar{\mathbf{w}} = \frac{\mathbf{w}}{w_1} \tag{2}$$

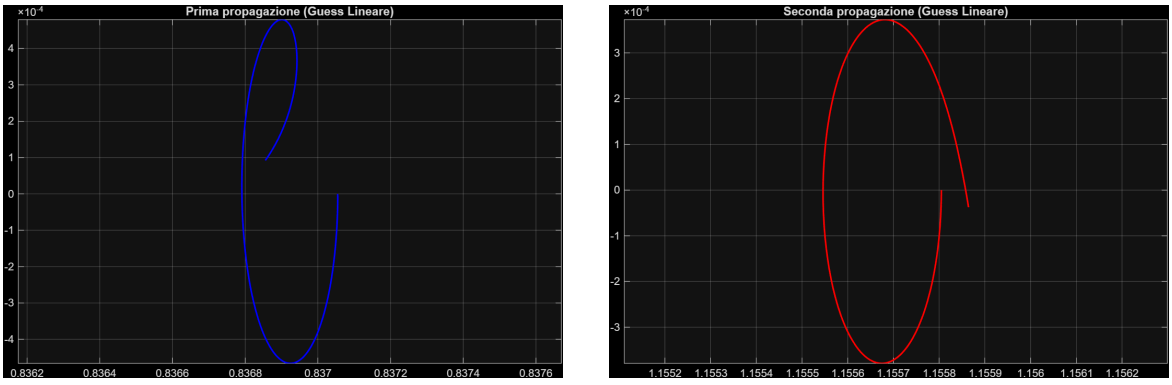
$$T_{guess} = \frac{2\pi}{\omega} \tag{3}$$

$$\mathbf{X}_{0guess} = \mathbf{X}_{L_1} + \alpha \cdot \bar{\mathbf{w}} \tag{4}$$

where:

- λ is the complex eigenvalue
- w is the natural frequency, used to normalize the eigenvector associated with the chosen eigenvalue
- T_{guess} is the estimate of the linear period of the orbit
- α is the orbit amplitude, the initial displacement from the equilibrium point fixed at 50km (normalized as 50/384400)

For the propagation ODE113 is used, with the dynamical model previously described. The same process is used to evaluate the orbit around both L1 and L2, but due to the non-linearity of the problem, the linear guess is not precise enough for the orbit to close.



(a) Linear Guess L1 (b) Linear Guess L2

Figure 2: Propagation of linear guesses

From these orbit plots, it's clear that the initial guess is not precise enough to close the periodic orbit, to avoid this problem a differential correction has been performed, the Single Shooting.

2.3 Single Shooting correction

To obtain a closed periodic orbit it's necessary to correct the initial state, considering the nonlinearity of the problem. The differential corrector is based on the Newton method, an iterative process with the goal of obtaining a final gap distance small enough ($gap < 10^{-12}$)

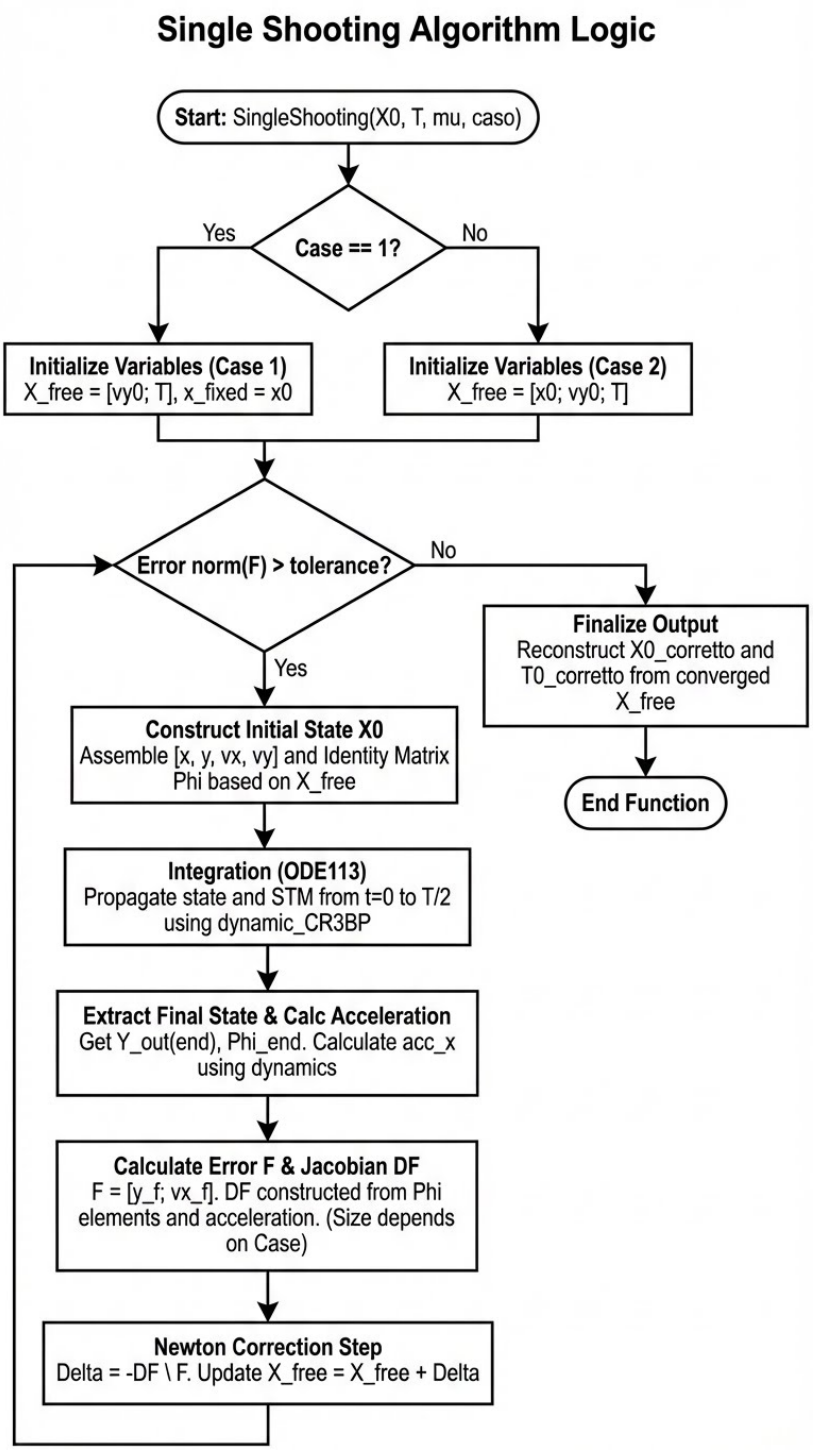


Figure 3: Single shooting algorithm

This algorithm is divided into two cases, to address this problem, case 2 is used, where x_0 is present inside the variable vector x_0, because we are looking for the best initial guess also in terms of x value.

$$\mathbf{X}_{\text{free}} = \begin{bmatrix} x_0 \\ \dot{y}_0 \\ T \end{bmatrix} \tag{5}$$

The error vector called F is represented by the value of y and u (or dx) and at the final time should be equal to zero, once this value is lower than the tolerance the loop breaks, meaning that the guess is accurate enough for the orbit to close.

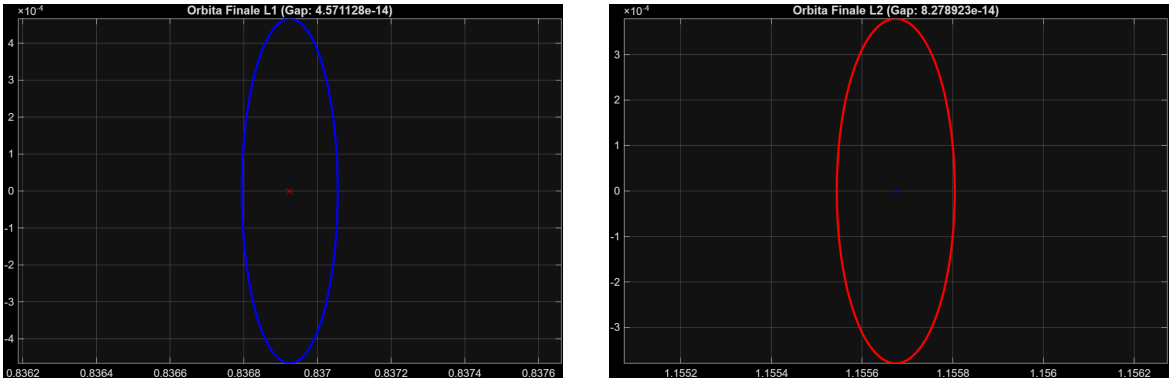
$$\mathbf{F}(\mathbf{X}_{\text{free}}) = \begin{bmatrix} y(T/2) \\ \dot{x}(T/2) \end{bmatrix} = \mathbf{0} \tag{6}$$

Each time the condition is not satisfied the guess is corrected by a value DF called Jacobian matrix of $\mathbf{F}$, that depends on phi and v (or dy) and du (acceleration along x).

$$\mathbf{X}_{\text{free}}^{(k+1)} = \mathbf{X}_{\text{free}}^{(k)} - DF^{-1}\mathbf{F}(\mathbf{X}_{\text{free}}^{(k)}) \tag{7}$$

$$DF = \begin{bmatrix} \frac{\partial y_f}{\partial x_0} & \frac{\partial y_f}{\partial \dot{y}_0} & \frac{\partial y_f}{\partial T} \\ \frac{\partial \dot{x}_f}{\partial x_0} & \frac{\partial \dot{x}_f}{\partial \dot{y}_0} & \frac{\partial \dot{x}_f}{\partial T} \end{bmatrix} = \begin{bmatrix} \phi_{2,1} & \phi_{2,4} & \frac{1}{2}\dot{y}_f \\ \phi_{3,1} & \phi_{3,4} & \frac{1}{2}\ddot{x}_f \end{bmatrix} \tag{8}$$

Once the algorithm ends the gap problem is solved, the correct guess now takes into account all the discrepancy of linearization allowing the orbit to close.



(a) Correct Guess L1 (b) Correct Guess L2

Figure 4: Propagation of correct guesses

2.4 Continuation scheme

Once the stable periodic orbit is reached, it's possible to compute the whole family of periodic orbits, because multiple solutions satisfy the conditions, these solutions differ by the initial state condition, in this case the displacement α. The family of orbits will be evaluated by increasing this value α by a quantity Δstep.

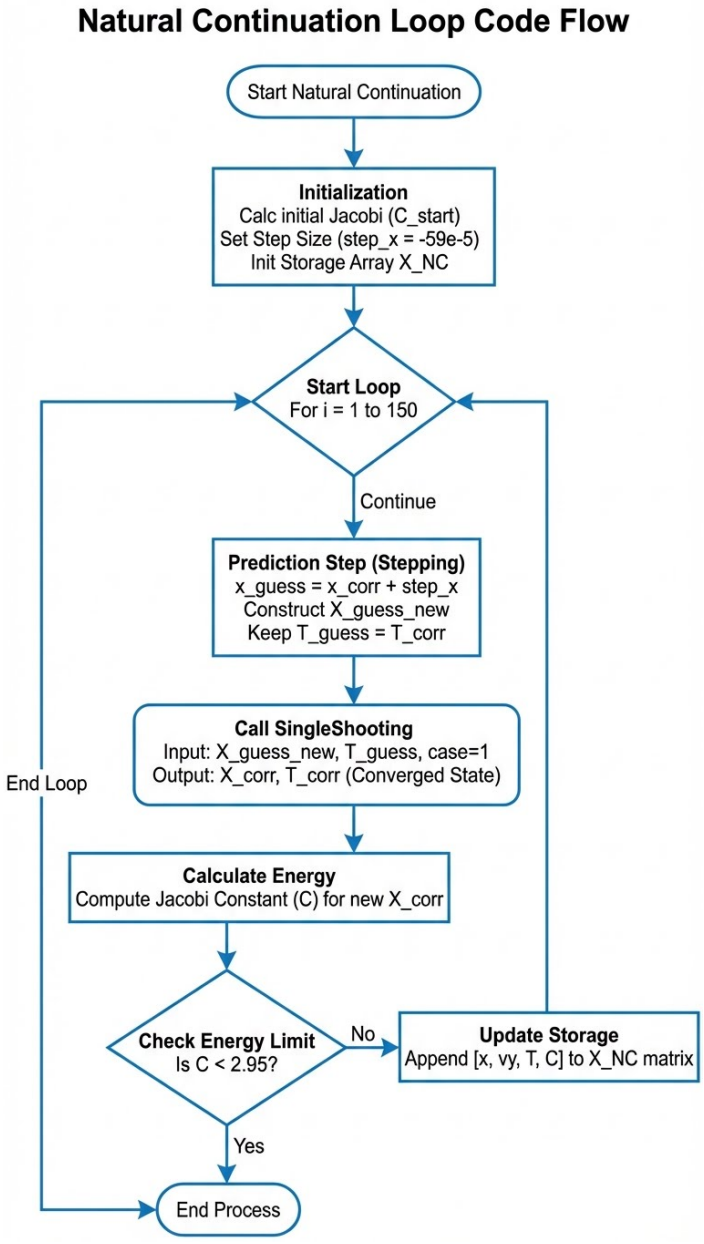


Figure 5: Natural continuation

The value of the step size, is found by iterating the process as the number of orbits, with the goal of obtaining a plot large enough to highlights the influence of the celestial bodies of the CR3BP, the Earth and the Moon. This time the single shooting function will not iterate the value of x, this is case 1, the value of x depends only on the step size, the single shooting is useful to find the correct guess of u and T. There's

a boundary condition on Jacobi constant, it can't go below the value of 2.95, because after this critical threshold the orbit is more likely to be unstable.

$$C(\mathbf{X}) = 2 \cdot U - (u^2 + v^2) \tag{9}$$

The same process has been done for both L1 and L2 family orbits, with different values of Δstep and different number of orbits, to obtain two families with similar extension, as shown on table below. It's interesting to notice the shape resembles a bean, mirrored for L2 family, this is caused by the presence of the Moon and its gravitational gradient, orbits that pass closer to the moon will lose their periodicity, so they will not be included in Lyapunov families.

Table 2: Different input of natural continuation

Parameter	**Value**
Δstep L1	$-59 \cdot 10^-5$
Δstep L2	$1.96 \cdot 10^-4$
N orbits L1	487
N orbits L2	139

These values come from numerous iterations, with the goal of obtaining a final orbit comparable to the case study. The orbit numbers stop once the Jacobi constant reaches a value lower than 2.95, because the orbit linked with this value will be too large so too unstable for our interest.

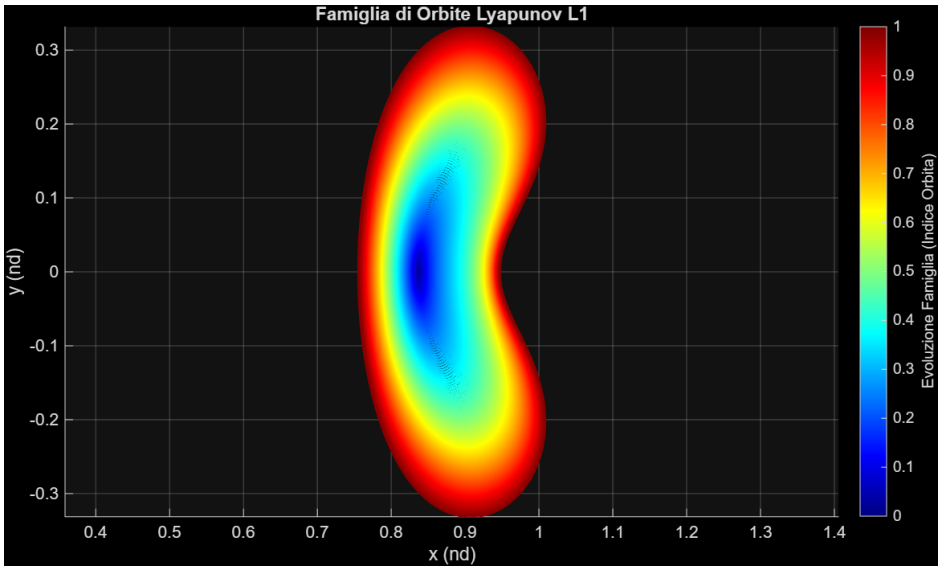


Figure 6: Family L1

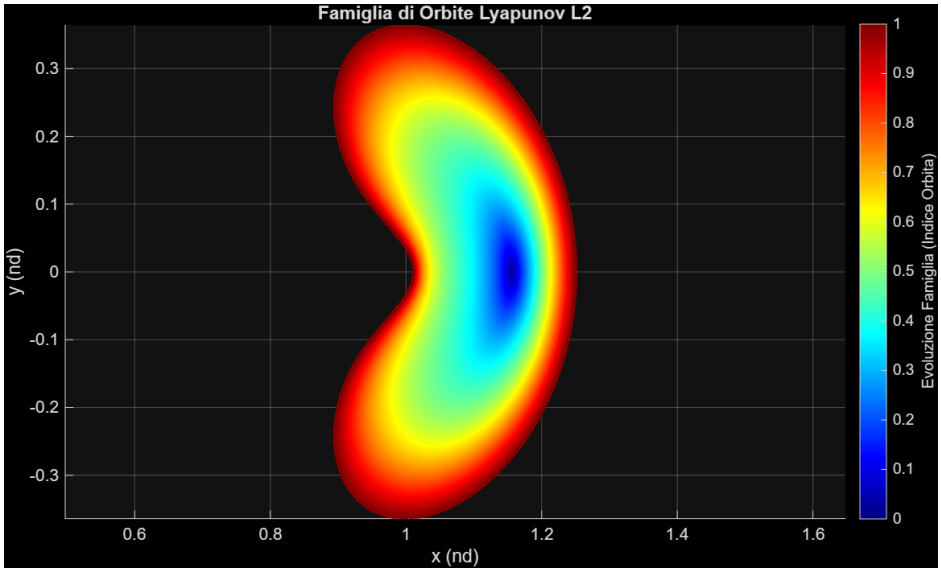


Figure 7: Family L2

The images above represent the orbits families, each orbit differs from the others in its family in terms of its energy state, the Jacobi constant represent this energy state, the lower the value the higher the energy state. It's interesting to notice that two orbits around different points can belong to the same energy level, this allow a hypothetical ballistic transfer between them.

2.5 Orbit pairs

To obtain two orbits with the same Jacobi constant value it's necessary to understand the bisection method. The natural continuation of the two families produced different orbits with different Jacobi constants, but it's almost impossible that two orbits (one around L1 and the other around L2) share this value, because of the discretization of the natural continuation. To solve this problem the bisection method is used, with a Master-Slave base, one orbit is selected from the family around L1, the only constraint is the Jacobi constant of the orbit that has to be inside a threshold [3.1370,3.1493]. Once the orbit is chosen, the next step is to evaluate its twin around L2, so an orbit with the same Jacobi constant, to do so the bisection method is used, as shown in the block diagram below. It's important to use the singleshooting function once the initial state of the orbit is found, because of the instability of the Lyapunov orbits

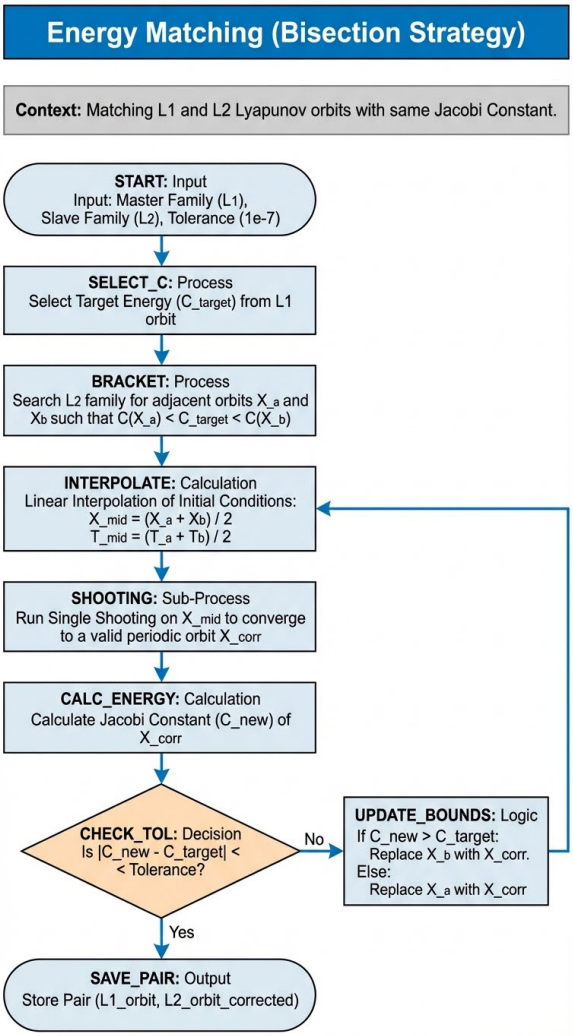


Figure 8: Bisection algorithm

This algorithm has been repeated for 4 orbits with 4 different values of Jacobi constant, these results will be useful for the evaluation of manifolds and ballistic

trajectories, so useful for no-thrust orbit changes.

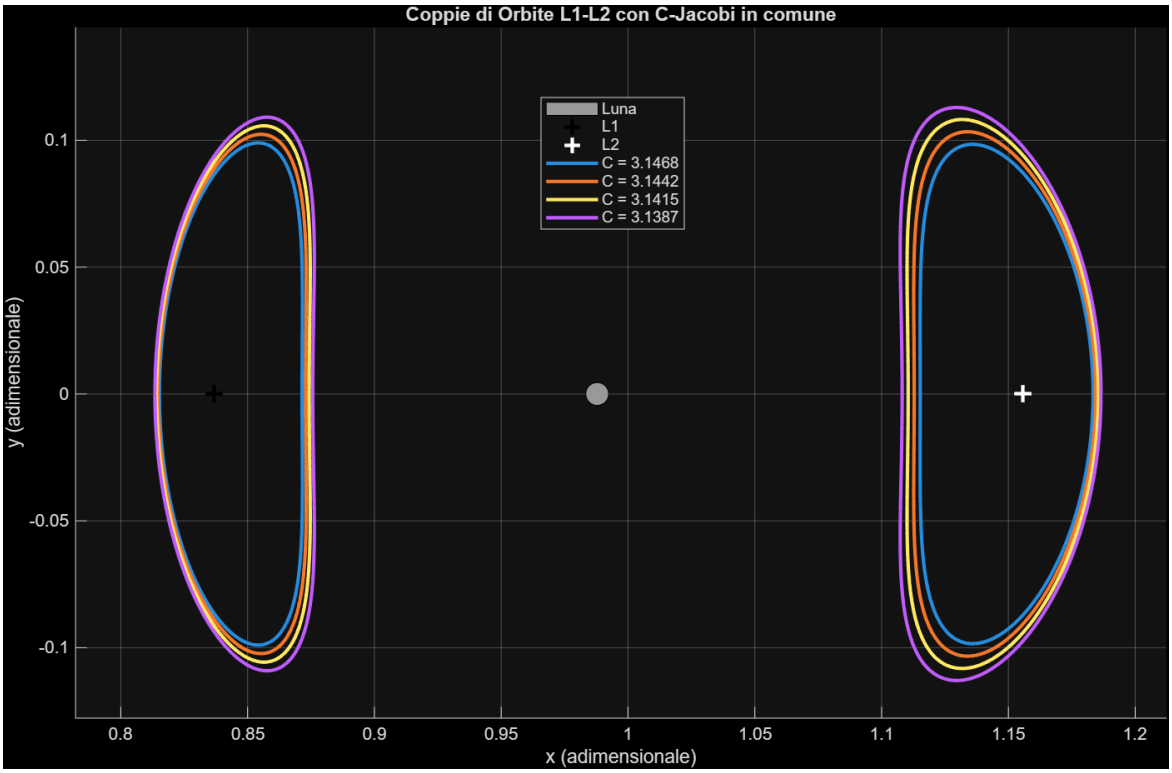


Figure 9: Orbit pairs

3 Full Algorithm

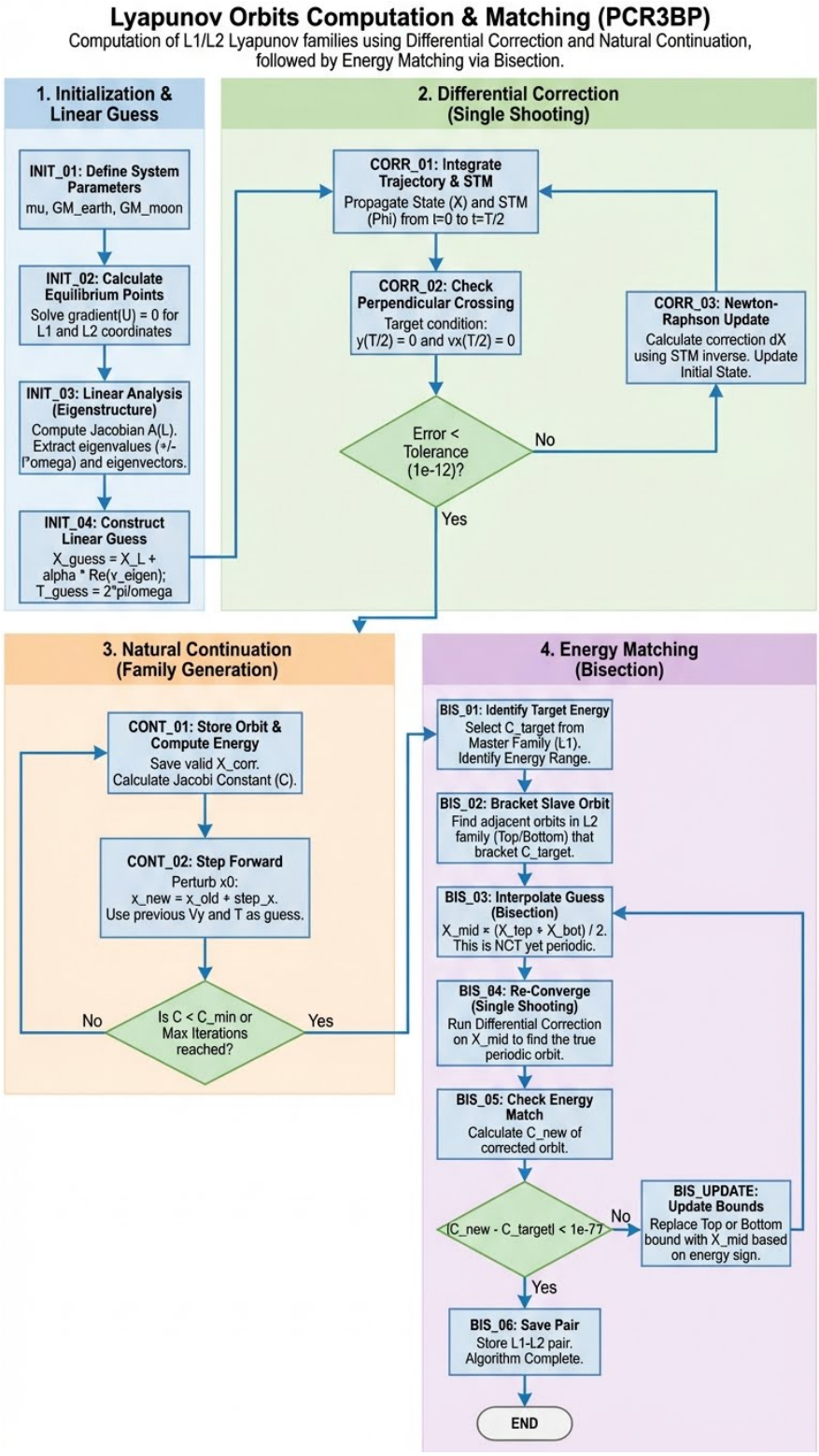


Figure 10: Full Algorithm